Mohnish Sahu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.76
Kendriya Vidyalaya Durg	Class XII (Central Board of Secondary Eduation)	2019	63.00%
Kendriya Vidyalaya Durg	Class X (Central Board of Secondary Eduation)	2017	10.00%

SKILLS

- Scientific Tools: MATLAB, Cadence, KiCAD - C2000 peripherals: ePWM, ADC, DAC, GPIOs, timers

- Software Platforms: Jupyter, CCSTUDIO - DSP: F280049C

- Programming Languages: C, C++, Python

RELEVANT COURSES

- Embedded System Design in Power Electronics - Power Converter Design - Power Electronics Lab
- Electrical Machines Analysis & Control - Multilevel Converters - Power Systems Lab

PROJECTS

PMSM Drive Control for EV Applications

- Modelling Interior **Permanent Magnet Synchronous Motor (IPMSM)** dynamics and implementing Field-Oriented Control (FOC) with MTPA-based torque optimization using MATLAB/Simulink.
- **Extending control design** into field-weakening and MTPV regions to enable high-speed operation, covering the full multi-region strategy (MTPA , FW , MTPV).
- **Studying performance evaluation** methods and hardware-oriented validation approaches for EV motor drive systems.

Modelling and closed-loop simulation of a 3-phase bi-directional VSC

- **Modelled 10 kVA bidirectional 3-phase 2-level VSC** with 800V DC link and 2 mH L-filter in MATLAB/Simulink.
- **Designed dq-frame PI current control** with pole-zero cancellation for unity-PF operation.
- **Implemented 35 kHz SPWM** and validated **±5 kW bidirectional power flow** between rectifier/inverter modes.

Closed-Loop Boost Converter Design & Hardware Implementation

- **Designed and built a working boost converter**, including inductor design, modelling, and control implementation.
- **Developed closed-loop voltage control** for different output-voltage references.
- **Compared open- and closed-loop performance** using MATLAB/Simulink simulations.

Real-Time Surge Voltage Monitoring & Protection System

- **Designed real-time surge protection** for industrial loads against overvoltage and undervoltage.
- **Implemented automatic load disconnection** using comparator, transistor and relay-based protection.
- **Integrated IoT-based monitoring** for remote voltage and fault-status tracking.

INTERSHIPS

Vocational Training -- NTPC, RLI Sipat

Report: Use of IoT in Electricity Sector

- Explored IoT applications in power generation, transmission, distribution, and consumption, including smart grids and asset monitoring.
- Studied Smart Grid and Smart Meter architecture, highlighting benefits over conventional power systems.
- Analyzed key IoT deployment challenges -- big data, connectivity, interoperability, and security -- in the power sector.

POSITIONS OF RESPONSIBILITY

Teaching Assistant (TA) -

- **Invigilated B.Tech tests**, ensuring smooth and fair conduct of assessments
- **Assisted in conducting B.Tech Machines Laboratory sessions**, supporting students during experiments and lab activities.

EXTRACURRICULAR

Zonal-Level Football Runner-Up (B.Tech) -- Represented the institute at the zonal-level tournament, securing 2nd place.

HOBBIES AND INTERESTS

Dancing, Playing Football, Following Football Leagues (UCL, La Liga, EPL)